

Alejandro Fontal, PhD

DATA SCIENCE | MACHINE LEARNING | BIOMEDICAL & HEALTH DATA

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Computational scientist with a PhD and 8+ years of academic and industry experience across health data, applied machine learning, epidemiology, and biomedical research. I develop statistical and ML methods and reproducible analyses for observational, experimental, and population-health data.

EXPERIENCE

NIMBLE Diagnostics

Senior Machine Learning Engineer

-  Jul. 2026 - Present 📍 Barcelona, ES
- Lead the machine learning strategy for a MedTech R&D program developing diagnostic methods from complex biomedical signals collected in experimental and clinical studies.
 - Develop statistical and ML approaches and validation workflows, working with signal-processing and clinical teams to link model outputs to clinically meaningful endpoints.

Climate & Health Program @ ISGlobal

Postdoctoral Researcher | Data & Software Lead

-  Oct. 2024 - Apr. 2026 📍 Barcelona, ES
- Led software and data engineering for the group, building internal packages, web tools, and GitHub-based workflows that supported reproducible research and team-wide development.
 - Built reproducible pipelines for long-read aerobiome metagenomics and linked results to health outcomes.
 - Developed supervised ML models for bacterial discrimination from laser-induced fluorescence and light-scattering measurements.

PhD Fellow | Early Stage Researcher (ESR)

-  Oct. 2019 - Oct. 2024
- Marie Skłodowska-Curie Actions ESR within HELICAL ITN, integrating health-registry, surveillance, and environmental data in international studies.*
- Developed time-series methods to quantify environmental signals in epidemics, including transient and lagged associations, multi-scale correlations, and phase-shift analyses.
 - Integrated nationwide case data with climate reanalysis, remote sensing, GIS layers, and air-mass back-trajectories to model drivers of Kawasaki Disease, COVID-19, and influenza.
 - Analyzed registry and emergency-health datasets including ANCA-associated vasculitis, AML alerts, and nationwide cardiac-arrest records, assessing data quality, denominators, and spatiotemporal incidence.

Protein Engineering @ DuPont Industrial Biosciences

Data Scientist

-  Sep. 2018 - Aug. 2019 📍 Leiden, NL
- Pioneered R&D efforts to integrate data-driven solutions into the protein engineering workflow, with a strong focus on ML applications.
 - Built and maintained reproducible bioinformatics and data pipelines.
 - Partnered with wet-lab teams to design experiments and close the model-experiment loop, accelerating workflow automation.

EDUCATION

PhD in Biotechnology 2019-2024 Universitat de Barcelona

- PhD focused on environmental determinants of disease onset, spatiotemporal modeling, and long-read aerobiome metagenomics.

MSc in Bioinformatics 2016-2018 Wageningen University & Research

- Minor in Data Science, cum laude distinction.
- Thesis on interpretable deep learning for protein subcellular location prediction

BSc in Biotechnology 2011-2015 Universitat de Barcelona

- Thesis at the VHIR Bioinformatics Unit on meta-analysis of transcriptomics tools.

Languages: Spanish & Catalan (native), English (C2)

TECHNICAL SKILLS

Python: Expert (9+ years); primary language. Package development, testing and maintenance. End-to-end DS/ML work: data wrangling, visualization, statistical modeling, machine learning, and app deployment (pandas, polars, statsmodels, scikit-learn, tensorflow, pytorch, plotnine, fastapi).

R: Tidyverse, Bioconductor, statistical and epidemiological analysis, visualization, reproducible reports, and R/Python workflows.

Git & DevOps: Collaborative development, code review, and automated testing/deployment (GH Actions, GitLab CI/CD).

Web / APIs: Working knowledge of modern web development; REST services, deployment and hosting of static and lightweight web apps.

AI/Agentic Development: Practical experience with agentic coding and automation workflows (Codex, Claude Code, OpenCode).

Statistical & health data analysis: Time-series and spatiotemporal analysis, lagged associations, incidence estimation, population denominators, health registries, surveillance data, and model validation.

Scientific computing: Reproducible Python/R workflows with Git, Linux/HPC, containers, SLURM, Nextflow, and CI/CD.